STATISTICS IS THE GRAMMAR OF SCIENCE

PROBABILITY AND STATISTICS

LECTURE # 24

STATISTICAL INFERENCE

INFERENCE ABOUT DIFFERENCE OF MEANS

PREPARED BY
HAZBER SAMSON
FAST-NUCES ISLAMABAD

INFERENCE ABOUT TWO POPULATIONS

INFERENCE ABOUT DIFFERENCE OF POPULATION MEANS WHEN SIGMAS ARE KNOWN

CASE-1 *If* $H_0 : \mu_1 - \mu_2 = \mu_0$ *vs* $H_1 : \mu_1 - \mu_2 \neq \mu_0$

- When σ_1 *and* σ_2 are known then we use **test statistic**

$$Z = \frac{\left(\overline{x_1} - \overline{x_2}\right) - \left(\mu_1 - \mu_2\right)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

- When σ_1 *and* σ_2 are known then we use **P-Value**

$$P = 2[1 - \Phi(|z_0|)]$$

- The $(1-\alpha)$ 100% **confidence interval** for $\mu_1 - \mu_2$ is

$$\left(\overline{x_1} - \overline{x_2}\right) - Z_{\frac{\alpha}{2}} \cdot \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \leq \mu_1 - \mu_2 \leq \left(\overline{x_1} - \overline{x_2}\right) - Z_{\frac{\alpha}{2}} \cdot \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

CASE-2 *If* $H_0 : \mu_1 - \mu_2 \geq \mu_0$ *vs* $H_1 : \mu_1 - \mu_2 < \mu_0$

- When σ_1 *and* σ_2 are known then we use **test statistic**

$$Z = \frac{\left(\overline{x_1} - \overline{x_2}\right) - \left(\mu_1 - \mu_2\right)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

- When σ_1 *and* σ_2 are known then we use **P-Value**

$$P = \Phi(z_0)$$

- The $(1-\alpha)$ 100% **confidence interval** for μ is

$$\mu_1 - \mu_2 \leq \left(\overline{x_1} - \overline{x_2}\right) + Z_{\alpha} \cdot \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

CASE-3 *If* $H_0 : \mu_1 - \mu_2 \leq \mu_0$ *vs* $H_1 : \mu_1 - \mu_2 > \mu_0$

- When σ_1 *and* σ_2 are known then we use **test statistic**

$$Z = \frac{\left(\overline{x_1} - \overline{x_2}\right) - \left(\mu_1 - \mu_2\right)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

- When σ_1 *and* σ_2 are known then we use **P-Value**

$$P = 1 - \Phi(z_0)$$

- The $(1-\alpha)$ 100% **confidence interval** for μ is

$$\mu_1 - \mu_2 \geq \left(\overline{x_1} - \overline{x_2}\right) - Z_{\alpha} \cdot \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

HAZBER SAMSON

Z-TEST

If $H_\circ : (=, \leq, \geq)$ then $H_1 : (\neq, >, <)$

CRITICAL REGIONS IN Z-TEST

- If H_1 *contains* $\neq$ then C.R is $|z| \geq Z_{\frac{\alpha}{2}}$ (Two Sided)
- If H_1 *contains* $>$ then C.R is $Z > Z_\alpha$ (One Sided)
- If H_1 *contains* $<$ then C.R is $Z < -Z_\alpha$ (One Sided)

FREQUENTLY USED CRITICAL VALUES OF Z

Significance Level (α)	Confidence Level $(1-\alpha)$	One-Tailed Test	Two-Tailed Test
$\alpha = 0.01$	99%	$Z_\alpha = 2.33$	$Z_{\alpha/2} = 2.58$
$\alpha = 0.02$	98%	$Z_\alpha = 2.05$	$Z_{\alpha/2} = 2.33$
$\alpha = 0.03$	97%	$Z_\alpha = 1.88$	$Z_{\alpha/2} = 2.17$
$\alpha = 0.04$	96%	$Z_\alpha = 1.75$	$Z_{\alpha/2} = 2.05$
$\alpha = 0.05$	95%	$Z_\alpha = 1.65$	$Z_{\alpha/2} = 1.96$
$\alpha = 0.06$	94%	$Z_\alpha = 1.55$	$Z_{\alpha/2} = 1.88$
$\alpha = 0.07$	93%	$Z_\alpha = 1.48$	$Z_{\alpha/2} = 1.81$
$\alpha = 0.08$	92%	$Z_\alpha = 1.41$	$Z_{\alpha/2} = 1.75$
$\alpha = 0.09$	91%	$Z_\alpha = 1.34$	$Z_{\alpha/2} = 1.70$
$\alpha = 0.10$	90%	$Z_\alpha = 1.28$	$Z_{\alpha/2} = 1.65$

P-VALUE FORMULAS IN Z-TEST

$$P = \begin{cases} 2[1-\Phi(|z_0|)] & \text{if} \quad H_1 : \mu \neq \mu_0 \\ \Phi(z_0) & \text{if} \quad H_1 : \mu < \mu_0 \\ 1-\Phi(z_0) & \text{if} \quad H_1 : \mu > \mu_0 \end{cases}$$

HAZBER SAMSON

EXAMPLES OF INFERENCE ABOUT DIFFERENCE OF MEANS

EXAMPLE-1 A survey found that the average hotel room rate in New Orleans is \$88.42 and the average room rate in Phoenix is \$80.61. Assume that the data were obtained from two samples of 50 hotels each and that the standard deviations of the populations are \$5.62 and \$4.83, respectively. At a 0.05, can it be concluded that there is a significant difference in the rates?

SOLUTION *Here* $n_1 = 50,\ n_2 = 50,\ \overline{x_1} = 88.42,\ \overline{x_2} = 80.61,\ \sigma_1 = 5.62, \sigma_2 = 4.83,\ \alpha = 0.05$

CRITICAL VALUE APPROACH

Step -1 : *Formulation of Hypotheses*

$H_0 : \mu_1 = \mu_2$

$H_1 : \mu_1 \neq \mu_2$

Step -2 : *Level of Significance*

$\alpha = 0.05$

Step -3 : *Test Statistic*

$$Z = \frac{\left(\overline{x_1} - \overline{x_2}\right) - \left(\mu_1 - \mu_2\right)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{\left(88.42 - 80.61\right) - \left(0\right)}{\sqrt{\frac{5.62^2}{50} + \frac{4.83^2}{50}}} = 7.45$$

Step -4 : *Critical* Re *gion*

$$|z| \geq z_{\frac{\alpha}{2}} \quad \Rightarrow |z| \geq 1.96$$

Step -5 : *Conclusion*

Since calculated value of z *lies in CR so reject* H_0.

P-VALUE APPROACH

Step -1 : *Formulation of Hypotheses*

$H_0 : \mu_1 = \mu_2$

$H_1 : \mu \neq \mu_2$

Step -2 : *Level of Significance*

$\alpha = 0.05$

Step -3 : *Test Statistic*

$$Z = \frac{\left(\overline{x_1} - \overline{x_2}\right) - \left(\mu_1 - \mu_2\right)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{\left(88.42 - 80.61\right) - \left(0\right)}{\sqrt{\frac{5.62^2}{50} + \frac{4.83^2}{50}}} = 7.45$$

Step -4 : $P - value$

$$p = 2[1 - \Phi(|z_0|)] = 2[1 - \Phi(|7.45|)] = 0.0002$$

Step -5 : *Conclusion*

Since $p - value < \alpha$ *so reject* H_0.

CONFIDENCE INTERVAL APPROACH

Step -1 : *Formulation of Confidence Interval*

The $(1-\alpha)100\%$ *CI for* $\mu_1 - \mu_2$ *is given by*

$$\left(\overline{x_1} - \overline{x_2}\right) - Z_{\frac{\alpha}{2}} \cdot \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \leq \mu_1 - \mu_2 \leq \left(\overline{x_1} - \overline{x_2}\right) + Z_{\frac{\alpha}{2}} \cdot \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

$$\left(88.42 - 80.61\right) - (1.96)\sqrt{\frac{5.62^2}{50} + \frac{4.83^2}{50}} \leq \mu_1 - \mu_2 \leq \left(88.42 - 80.61\right) + (1.96)\sqrt{\frac{5.62^2}{50} + \frac{4.83^2}{50}}$$

$$5.76 \leq \mu \leq 9.86$$

Step -2 : *Conclusion*

Since $\mu_1 - \mu_2 = 0$ *does not lie in CI so reject* H_0.

HAZBER SAMSON

EXAMPLE-2 A local college cafeteria has a self-service soft ice cream machine. The cafeteria provides bowls that can hold up to 16 ounces of ice cream. The food service manager is interested in comparing the average amount of ice cream dispensed by male students to the average amount dispensed by female students. A measurement device was placed on the ice cream machine to determine the amounts dispensed. Random samples of 85 male and 78 female students who got ice cream were selected. The sample averages were 7.23 and 6.49 ounces for the male and female students, respectively. Assume that the population standard deviations are 1.22 and 1.17 ounces, respectively. Using a 1% significance level, can you conclude that the average amount of ice cream dispensed by all male college students is larger than the average amount dispensed by all female college students?

SOLUTION *Here* $n_1 = 85,\ n_2 = 78,\ \overline{x_1} = 7.23,\ \overline{x_2} = 6.49,\ \sigma_1 = 1.22,\ \sigma_2 = 1.17,\ \alpha = 0.01$

CRITICAL VALUE APPROACH

Step -1 : *Formulation of Hypotheses*

$H_0 : \mu_1 \le \mu_2$

$H_1 : \mu_1 > \mu_2$

Step -2 : *Level of Significance*

$\alpha = 0.01$

Step -3 : *Test Statistic*

$$Z = Z = \frac{\left(\overline{x_1} - \overline{x_2}\right) - \left(\mu_1 - \mu_2\right)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{(7.23 - 6.49) - (0)}{\sqrt{\frac{1.22^2}{85} + \frac{1.17^2}{78}}} = \frac{0.74}{0.1872} = 3.95$$

Step -4 : *Critical* Re *gion*

$z > z_\alpha \quad \Rightarrow \ z > z_{0.01} \quad \Rightarrow z > 2.33$

Step -5 : *Conclusion*

Since calculated value of z *lies in CR so reject* H_0.

P-VALUE APPROACH

Step -1 : *Formulation of Hypotheses*

$H_0 : \mu_1 \le \mu_2$

$H_1 : \mu_1 > \mu_2$

Step -2 : *Level of Significance*

$\alpha = 0.01$

Step -3 : *Test Statistic*

$$Z = Z = \frac{\left(\overline{x_1} - \overline{x_2}\right) - \left(\mu_1 - \mu_2\right)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{(7.23 - 6.49) - (0)}{\sqrt{\frac{1.22^2}{85} + \frac{1.17^2}{78}}} = \frac{0.74}{0.1872} = 3.95$$

Step -4 : *P* $-$ *value*

$p = 1 - \Phi(z_0) = 1 - \Phi(3.95) = 1 - 0.9999 = 0.0001$

Step -5 : *Conclusion*

Since $p -$ *value* $< \alpha$ *so reject* H_0.

CONFIDENCE INTERVAL APPROACH

Step -1 : *Formulation of Confidence Interval*

The $(1-\alpha)100\%$ *CI for* $\mu_1 - \mu_2$ *is given by*

$$\mu_1 - \mu_2 \ge \left(\overline{x_1} - \overline{x_2}\right) - Z_\alpha . \sqrt{\frac{\sigma_1^{\ 2}}{n_1} + \frac{\sigma_2^{\ 2}}{n_2}}$$

$$\mu_1 - \mu_2 \ge (7.23 - 6.49) - (2.33)\sqrt{\frac{1.22^2}{85} + \frac{1.17^2}{78}}$$

$$\mu_1 - \mu_2 \ge 0.3038$$

Step -2 : *Conclusion*

Since $\mu_1 - \mu_2 = 0$ *does not lie in CI so reject* H_0.

HAZBER SAMSON

EXAMPLE-3 **REM (rapid eye movement)** sleep is sleep is during which most dreams occur. Each night a person has both REM and non-REM sleep. However it is thought that adults have less REM sleep than children. Assume that REM sleep time is normally distributed for both adults and children. A random sample of 10 adults showed that they have average REM sleep time of 2.1 hours per night with population standard deviation 0.7 hour. Another random sample of 10 children showed that they have average REM sleep time of 2.8 hours per night with population standard deviation 0.5 hour. Do the data indicate, on average, adults tend to have less REM sleep than children? Use 1% level of significance.

SOLUTION *Here* $n_1 = 10,\ n_2 = 10,\ \overline{x_1} = 2.1,\ \overline{x_2} = 2.8,\ \sigma_1 = 0.7, \sigma_2 = 0.5,\ \alpha = 0.01$

CRITICAL VALUE APPROACH

Step -1: *Formulation of Hypotheses*

$H_0 : \mu_1 \geq \mu_2$

$H_1 : \mu_1 < \mu_2$

Step -2: *Level of Significance*

$\alpha = 0.01$

Step -3: *Test Statistic*

$$Z = \frac{\left(\overline{x_1} - \overline{x_2}\right) - \left(\mu_1 - \mu_2\right)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{\left(2.1 - 2.8\right) - \left(0\right)}{\sqrt{\frac{0.7^2}{10} + \frac{0.5^2}{10}}} = \frac{-0.7}{0.2720} = -2.57$$

Step -4: *Critical Region*

$z < -z_\alpha \quad \Rightarrow\ z < -2.33$

Step -5: *Conclusion*

Since calculated value of z *lies in CR so reject* H_0.

P-VALUE APPROACH

Step -1: *Formulation of Hypotheses*

$H_0 : \mu_1 \geq \mu_2$

$H_1 : \mu_1 < \mu_2$

Step -2: *Level of Significance*

$\alpha = 0.01$

Step -3: *Test Statistic*

$$Z = \frac{\left(\overline{x_1} - \overline{x_2}\right) - \left(\mu_1 - \mu_2\right)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{\left(2.1 - 2.8\right) - \left(0\right)}{\sqrt{\frac{0.7^2}{10} + \frac{0.5^2}{10}}} = \frac{-0.7}{0.2720} = -2.57$$

Step -4: $P-$*value*

$p = \Phi\left(z_0\right) = \Phi\left(-2.57\right) = 0.0051$

Step -5: *Conclusion*

Since $p-$*value* $< \alpha$ *so reject* H_0.

CONFIDENCE INTERVAL APPROACH

Step -1: *Formulation of Confidence Interval*

The $(1-\alpha)100\%$ *CI for* $\mu_1 - \mu_2$ *is given by*

$$\mu_1 - \mu_2 \leq \left(\overline{x_1} - \overline{x_2}\right) + Z_\alpha \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

$$\mu_1 - \mu_2 \leq \left(2.1 - 2.8\right) + (2.33)\sqrt{\frac{0.7^2}{10} + \frac{0.5^2}{10}}$$

$$\mu_1 - \mu_2 \leq -1.33$$

Step -2: *Conclusion*

Since $\mu_1 - \mu_2 = 0$ *does not lie in CI so reject* H_0.

HAZBER SAMSON

INFERENCE ABOUT TWO POPULATIONS

INFERENCE ABOUT DIFFERENCE OF POPULATION MEANS WHEN SIGMAS ARE UNKNOWN

CASE-1 *If* $H_0 : \mu_1 - \mu_2 = \mu_0$ *vs* $H_1 : \mu_1 - \mu_2 \neq \mu_0$

- When σ_1 *and* σ_2 are unknown then we use **test statistic**

$$t = \frac{(\overline{x_1} - \overline{x_2}) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad with \quad d.f = n_1 + n_2 - 2 \; where \; S_p = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}}$$

- When σ_1 *and* σ_2 are unknown then we use **P-Value**

$$P = 2 \times P(X \geq |t_0|)$$

- The $(1-\alpha)$ 100% **confidence interval** for $\mu_1 - \mu_2$ is

$$(\overline{x_1} - \overline{x_2}) - t_{\frac{\alpha}{2}(d.f)} S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \leq \mu_1 - \mu_2 \leq (\overline{x_1} - \overline{x_2}) + t_{\frac{\alpha}{2}(d.f)} S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

CASE-2 *If* $H_0 : \mu_1 - \mu_2 \geq \mu_0$ *vs* $H_1 : \mu_1 - \mu_2 < \mu_0$

- When σ_1 *and* σ_2 are unknown then we use **test statistic**

$$t = \frac{(\overline{x_1} - \overline{x_2}) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad with \quad d.f = n_1 + n_2 - 2 \; where \; S_p = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}}$$

- When σ_1 *and* σ_2 are unknown then we use **P-Value**

$$P = P(X \leq t_0)$$

- The $(1-\alpha)$ 100% **confidence interval** for μ is

$$\mu_1 - \mu_2 \leq (\overline{x_1} - \overline{x_2}) + t_{\alpha(d.f)} S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

CASE-3 *If* $H_0 : \mu_1 - \mu_2 \leq \mu_0$ *vs* $H_1 : \mu_1 - \mu_2 > \mu_0$

- When σ_1 *and* σ_2 are unknown then we use **test statistic**

$$t = \frac{(\overline{x_1} - \overline{x_2}) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad with \quad d.f = n_1 + n_2 - 2 \; where \; S_p = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}}$$

- When σ_1 *and* σ_2 are unknown then we use **P-Value**

$$P = P(X \geq t_0)$$

- The $(1-\alpha)$ 100% **confidence interval** for μ is

$$\mu_1 - \mu_2 \geq (\overline{x_1} - \overline{x_2}) - t_{\frac{\alpha}{2}(d.f)} S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

HAZBER SAMSON

T-TEST

If $H_\circ : (=, \leq, \geq)$ then $H_1 : (\neq, >, <)$

Now for Critical Region

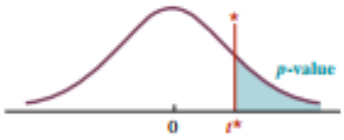
- If H_1 *contains* $\neq$ then C.R is $|t| \geq t_{\frac{\alpha}{2}(n_1+n_2-2)}$ (Two Sided)
- If H_1 *contains* $>$ then C.R is $t > t_{\alpha(n_1+n_2-2)}$ (One Sided)
- If H_1 *contains* $<$ then C.R is $t < -t_{\alpha(n_1+n_2-2)}$ (One Sided)

STUDENTS T-DISTRIBUTION TABLE

df	α = 0.1	0.05	0.025	0.01	0.005	0.001	0.0005
1	3.078	6.314	12.706	31.821	63.656	318.289	636.578
2	1.886	2.92	4.303	6.965	9.925	22.328	31.6
3	1.638	2.353	3.182	4.541	5.841	10.214	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.61
5	1.476	2.015	2.571	3.365	4.032	5.894	6.869
6	1.44	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.86	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.25	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.93	4.318
13	1.35	1.771	2.16	2.65	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.14
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	1.337	1.746	2.12	2.583	2.921	3.686	4.015
17	1.333	1.74	2.11	2.567	2.898	3.646	3.965
18	1.33	1.734	2.101	2.552	2.878	3.61	3.922
19	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	1.325	1.725	2.086	2.528	2.845	3.552	3.85
21	1.323	1.721	2.08	2.518	2.831	3.527	3.819
22	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	1.319	1.714	2.069	2.5	2.807	3.485	3.768
24	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	1.316	1.708	2.06	2.485	2.787	3.45	3.725
26	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	1.314	1.703	2.052	2.473	2.771	3.421	3.689
28	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	1.311	1.699	2.045	2.462	2.756	3.396	3.66
30	1.31	1.697	2.042	2.457	2.75	3.385	3.646
60	1.296	1.671	2	2.39	2.66	3.232	3.46
120	1.289	1.658	1.98	2.358	2.617	3.16	3.373
∞	1.282	1.645	1.96	2.326	2.576	3.091	3.291

Probability-Values for Student's *t*-distribution

The entries in this table are the *p*-values related to the right-hand tail for the calculated $t\star$ value for the *t*-distribution of df degrees of freedom.



	Degrees of Freedom														
$t\star$	3	4	5	6	7	8	10	12	15	18	21	25	29	35	df ≥ 45
0.0	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500
0.1	0.463	0.463	0.462	0.462	0.462	0.461	0.461	0.461	0.461	0.461	0.461	0.461	0.461	0.460	0.460
0.2	0.427	0.426	0.425	0.424	0.424	0.423	0.423	0.422	0.422	0.422	0.422	0.422	0.421	0.421	0.421
0.3	0.392	0.390	0.388	0.387	0.386	0.386	0.385	0.385	0.384	0.384	0.384	0.383	0.383	0.383	0.383
0.4	0.358	0.355	0.353	0.352	0.351	0.350	0.349	0.348	0.347	0.347	0.347	0.346	0.346	0.346	0.346
0.5	0.326	0.322	0.319	0.317	0.316	0.315	0.314	0.313	0.312	0.312	0.311	0.311	0.310	0.310	0.310
0.6	0.295	0.290	0.287	0.285	0.284	0.283	0.281	0.280	0.279	0.278	0.277	0.277	0.277	0.276	0.276
0.7	0.267	0.261	0.258	0.255	0.253	0.252	0.250	0.249	0.247	0.246	0.246	0.245	0.245	0.244	0.244
0.8	0.241	0.234	0.230	0.227	0.225	0.223	0.221	0.220	0.218	0.217	0.216	0.216	0.215	0.215	0.214
0.9	0.217	0.210	0.205	0.201	0.199	0.197	0.195	0.193	0.191	0.190	0.189	0.188	0.188	0.187	0.186
1.0	0.196	0.187	0.182	0.178	0.175	0.173	0.170	0.169	0.167	0.165	0.164	0.163	0.163	0.162	0.161
1.1	0.176	0.167	0.161	0.157	0.154	0.152	0.149	0.146	0.144	0.143	0.142	0.141	0.140	0.139	0.139
1.2	0.158	0.148	0.142	0.138	0.135	0.132	0.129	0.127	0.124	0.123	0.122	0.121	0.120	0.119	0.118
1.3	0.142	0.132	0.125	0.121	0.117	0.115	0.111	0.109	0.107	0.105	0.104	0.103	0.102	0.101	0.100
1.4	0.128	0.117	0.110	0.106	0.102	0.100	0.096	0.093	0.091	0.089	0.088	0.087	0.086	0.085	0.084
1.5	0.115	0.104	0.097	0.092	0.089	0.086	0.082	0.080	0.077	0.075	0.074	0.073	0.072	0.071	0.070
1.6	0.104	0.092	0.085	0.080	0.077	0.074	0.070	0.068	0.065	0.064	0.062	0.061	0.060	0.059	0.058
1.7	0.094	0.082	0.075	0.070	0.066	0.064	0.060	0.057	0.055	0.053	0.052	0.051	0.050	0.049	0.048
1.8	0.085	0.073	0.066	0.061	0.057	0.055	0.051	0.049	0.046	0.044	0.043	0.042	0.041	0.040	0.039
1.9	0.077	0.065	0.058	0.053	0.050	0.047	0.043	0.041	0.038	0.037	0.036	0.035	0.034	0.033	0.032
2.0	0.070	0.058	0.051	0.046	0.043	0.040	0.037	0.034	0.032	0.030	0.029	0.028	0.027	0.027	0.026
2.1	0.063	0.052	0.045	0.040	0.037	0.034	0.031	0.029	0.027	0.025	0.024	0.023	0.022	0.022	0.021
2.2	0.058	0.046	0.040	0.035	0.032	0.029	0.026	0.024	0.022	0.021	0.020	0.019	0.018	0.017	0.016
2.3	0.052	0.041	0.035	0.031	0.027	0.025	0.022	0.020	0.018	0.017	0.016	0.015	0.014	0.014	0.013
2.4	0.048	0.037	0.031	0.027	0.024	0.022	0.019	0.017	0.015	0.014	0.013	0.012	0.012	0.011	0.010
2.5	0.044	0.033	0.027	0.023	0.020	0.018	0.016	0.014	0.012	0.011	0.010	0.010	0.009	0.009	0.008
2.6	0.040	0.030	0.024	0.020	0.018	0.016	0.013	0.012	0.010	0.009	0.008	0.008	0.007	0.007	0.006
2.7	0.037	0.027	0.021	0.018	0.015	0.014	0.011	0.010	0.008	0.007	0.007	0.006	0.006	0.005	0.005
2.8	0.034	0.024	0.019	0.016	0.013	0.012	0.009	0.008	0.007	0.006	0.005	0.005	0.005	0.004	0.004
2.9	0.031	0.022	0.017	0.014	0.011	0.010	0.008	0.007	0.005	0.005	0.004	0.004	0.004	0.003	0.003
3.0	0.029	0.020	0.015	0.012	0.010	0.009	0.007	0.006	0.004	0.004	0.003	0.003	0.003	0.002	0.002
3.1	0.027	0.018	0.013	0.011	0.009	0.007	0.006	0.005	0.004	0.003	0.003	0.002	0.002	0.002	0.002
3.2	0.025	0.016	0.012	0.009	0.008	0.006	0.005	0.004	0.003	0.002	0.002	0.002	0.002	0.001	0.001
3.3	0.023	0.015	0.011	0.008	0.007	0.005	0.004	0.003	0.002	0.002	0.002	0.001	0.001	0.001	0.001
3.4	0.021	0.014	0.010	0.007	0.006	0.005	0.003	0.003	0.002	0.002	0.001	0.001	0.001	0.001	0.001
3.5	0.020	0.012	0.009	0.006	0.005	0.004	0.003	0.002	0.002	0.001	0.001	0.001	0.001	0.001	0.001
3.6	0.018	0.011	0.008	0.006	0.004	0.004	0.002	0.002	0.001	0.001	0.001	0.001	0.001	0+	0+
3.7	0.017	0.010	0.007	0.005	0.004	0.003	0.002	0.002	0.001	0.001	0.001	0.001	0+	0+	0+
3.8	0.016	0.010	0.006	0.004	0.003	0.003	0.002	0.001	0.001	0.001	0.001	0+	0+	0+	0+
3.9	0.015	0.009	0.006	0.004	0.003	0.002	0.001	0.001	0.001	0.001	0+	0+	0+	0+	0+
4.0	0.014	0.008	0.005	0.004	0.003	0.002	0.001	0.001	0.001	0+	0+	0+	0+	0+	0+

For specific details about using this table to find *p*-values, see pages 484, 486.

P-VALUE FORMULAS IN T-TEST

$$P=\begin{cases} P=2\times P([\Phi(|t_0|)]) & if \quad H_1:\mu\neq\mu_0 \\ P=1-\Phi(|t_0|)\ if\ t_0>0\ \&\ P=\Phi(|t_0|)\ if\ t_0<0 & if \quad H_1:\mu<\mu_0 \\ P=\Phi(|t_0|)\ if\ t_0>0\ \&\ P=1-\Phi(|t_0|)\ if\ t_0<0 & if \quad H_1:\mu>\mu_0 \end{cases}$$

EXAMPLES OF INFERENCE ABOUT DIFFERENCE OF MEANS

EXAMPLE-1 A random sample of 10 observations from one population revealed a sample mean of 23 and a sample standard deviation of 4. A random sample of 8 observations from another population revealed a sample mean of 26 and a sample standard deviation of 5. Assuming equal variances, At the 0.05 significance level, is there a difference between the population means? Test the claim using

(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

SOLUTION *There are two Populations. Population* -1 *and Population* -2.

Here $n_1 = 10,\ \overline{x_1} = 23,\ S_1 = 4,\ n_2 = 8,\ \overline{x_2} = 26,\ S_2 = 5,\ \alpha = 0.05, df = 10+8-2 = 16$

$$S_p = \sqrt{\frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}} = \sqrt{\frac{(10-1)(4)^2 + (8-1)(5)^2}{10+8-2}} = \sqrt{\frac{319}{16}} = 4.46$$

CRITICAL VALUE APPROACH

Step -1: *Formulation of Hypotheses*

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

Step -2: *Level of Significance*

$$\alpha = 0.05$$

Step -3: *Test Statistic*

$$t = \frac{(\overline{x_1} - \overline{x_2}) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad with \quad d.f = n_1 + n_2 - 2 \text{ where } S_p = \sqrt{\frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}}$$

$$So \quad t = \frac{(\overline{x_1} - \overline{x_2}) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{(23-26)-(0)}{(4.46)\sqrt{\frac{1}{10} + \frac{1}{8}}} = \frac{-3}{2.1155} = -1.4181$$

Step -4: *Critical* Re *gion*

$$|t| \geq t_{\frac{\alpha}{2}(n_1+n_2-2)} \Rightarrow |t| \geq t_{\frac{0.05}{2}(10+8-2)} \Rightarrow |t| \geq t_{0.025(16)} \Rightarrow |t| \geq 2.12$$

Step -5: *Conclusion*

Since calculated value of t does not lie in CR so do not reject H_0.

P-VALUE APPROACH

Step -1: *Formulation of Hypotheses*

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

Step -2: *Level of Significance*

$$\alpha = 0.05$$

Step -3: *Test Statistic*

$$t = \frac{(\overline{x_1} - \overline{x_2}) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad with \quad d.f = n_1 + n_2 - 2 \text{ where } S_p = \sqrt{\frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}}$$

$$So \quad t = \frac{(\overline{x_1} - \overline{x_2}) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{(23-26)-(0)}{(4.46)\sqrt{\frac{1}{10} + \frac{1}{8}}} = \frac{-3}{2.1155} = -1.4181$$

Step -4: $P-$*value*

$$P = 2 \times P([\Phi(|t_0|)]$$

$$P = 2 \times P([\Phi(|-1.4|)] = 2 \times 0.091 = 0.182$$

Step -5: *Conclusion*

Since $P-value > \alpha$ *so do not reject* H_0

HAZBER SAMSON

CONFIDENCE INTERVAL APPROACH

Step -1: *Formulation of Confidence Interval*

The $(1-\alpha)100\%$ *CI for* $\mu_1-\mu_2$ *is given by*

$$(23-26)-t_{\frac{\alpha}{2}(d.f)}S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}} \le \mu_1-\mu_2 \le (\overline{x_1}-\overline{x_2})+t_{\frac{\alpha}{2}(d.f)}S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}$$

$$(-3)-(2.12)(4.46)\sqrt{\frac{1}{10}+\frac{1}{8}} \le \mu_1-\mu_2 \le (-3)-(2.12)(4.46)\sqrt{\frac{1}{10}+\frac{1}{8}}$$

$$-3-4.1718 \le \mu_1-\mu_2 \le -3-4.1718$$

$$-7.1817 \le \mu_1-\mu_2 \le 1.1718$$

Step -2: *Conclusion*

Since $\mu_1-\mu_2=0$ *lies in CI so do not reject* H_0.

EXAMPLE-2 The average mpg usage for a 2009 Toyota Prius for a sample of 10 tanks of gas was 45.5 with a standard deviation of 1.8. For a 2009 Honda Insight, the average mpg usage for a sample of 10 tanks of gas was 42.0 with a standard deviation of 2.3. Assuming equal variances, at $\alpha=0.01$, is the true mean mpg lower for the Toyota Prius? Assuming equal variances, Check the claim using (a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

SOLUTION Let Type-1 denote Toyota and Type-2 denote Honda.

Here $n_1=10,\ \overline{x_1}=45.5,\ S_1=1.8,\ n_2=10,\ \overline{x_2}=42,\ S_2=2.3,\ \alpha=0.01, df=10+10-2=18$

$$S_p=\sqrt{\frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2}}=\sqrt{\frac{(10-1)(1.8)^2+(10-1)(2.3)^2}{10+10-2}}=\sqrt{\frac{76.77}{18}}=2.06$$

CRITICAL VALUE APPROACH

Step -1: *Formulation of Hypotheses*

$H_0: \mu_1 \ge \mu_2$

$H_1: \mu_1 < \mu_2$

Step -2: *Level of Significance*

$\alpha=0.01$

Step -3: *Test Statistic*

$$t=\frac{(\overline{x_1}-\overline{x_2})-(\mu_1-\mu_2)}{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \quad with \quad d.f=n_1+n_2-2 \ where\ S_p=\sqrt{\frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2}}$$

$$So \quad t=\frac{(\overline{x_1}-\overline{x_2})-(\mu_1-\mu_2)}{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}}=\frac{(45.5-42)-(0)}{(2.06)\sqrt{\frac{1}{10}+\frac{1}{10}}}=\frac{3.5}{0.8574}=4.08$$

Step -4: *Critical* Re *gion*

$$t<-t_{\alpha(n_1+n_2-2)} \Rightarrow t<-t_{(0.01)(10+8-2)} \Rightarrow t<-t_{0.01(16)} \Rightarrow t<-2.583$$

Step -5: *Conclusion*

Since calculated value of t *does not lie in CR so do not reject* H_0.

HAZBER SAMSON

P-VALUE APPROACH

Step -1: *Formulation of Hypotheses*

$H_0 : \mu_1 \geq \mu_2$

$H_1 : \mu_1 < \mu_2$

Step -2: *Level of Significance*

$\alpha = 0.01$

Step -3: *Test Statistic*

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad with \quad d.f = n_1 + n_2 - 2 \text{ where } S_p = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}}$$

$$So \quad t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{(45.5 - 42) - (0)}{(2.06)\sqrt{\frac{1}{10} + \frac{1}{10}}} = \frac{3.5}{0.8574} = 4.08$$

Step -4: $P - value$

$p = 1 - \Phi(t_0) = 1 - \Phi(4.08) = 0.9995$

Step -5: *Conclusion*

Since $p - value > \alpha$ *so do not reject* H_0.

CONFIDENCE INTERVAL APPROACH

Step -1: *Formulation of Confidence Interval*

The $(1-\alpha)100\%$ *CI for* $\mu_1 - \mu_2$ *is given by*

$$\mu_1 - \mu_2 \leq (\bar{x}_1 - \bar{x}_2) + t_{\alpha(d.f)} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$\mu_1 - \mu_2 \leq (42.5 - 42) + t_{0.01(16)}(2.06)\sqrt{\frac{1}{10} + \frac{1}{10}}$$

$$\mu_1 - \mu_2 \leq (42.5 - 42) + (2.583)(2.06)\sqrt{\frac{1}{10} + \frac{1}{10}}$$

$$\mu_1 - \mu_2 \leq 5.88$$

Step -2: *Conclusion*

Since $\mu_1 - \mu_2 = 0$ *lies in CI so do not reject* H_0.

EXAMPLE-3 When the background music tempo was slow, the mean amount of bar purchases for a sample of 17 restaurant patrons was \$30.47 with a standard deviation of \$15.10. When the background music tempo was fast, the mean amount of bar purchases for a sample of 14 patrons in the same restaurant was \$21.62 with a standard deviation of \$9.50. (a) Assuming equal variances, at $\alpha = 0.01$, is the true mean higher when the music is slow? Check the claim using (a) *Critical value Approach* (b) $P - value$ *Approach* (c) *Confidence Interval Approach*

SOLUTION *Population* -1: *Slow Music Tempo,* *Population* -2: *Fast Music Tempo*

Here $n_1 = 17,\ \bar{x}_1 = 30.47,\ S_1 = 15.10,\ n_2 = 14,\ \bar{x}_2 = 21.62,\ S_2 = 9.50,\ \alpha = 0.01,\ df = 17 + 14 - 2 = 29$

$$S_p = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{(17 - 1)(15.10)^2 + (14 - 1)(9.50)^2}{17 + 14 - 2}} = \sqrt{\frac{4821.41}{29}} = 12.89$$

HAZBER SAMSON

CRITICAL VALUE APPROACH

Step -1: *Formulation of Hypotheses*

$H_0 : \mu_1 \le \mu_2$

$H_1 : \mu_1 > \mu_2$

Step -2: *Level of Significance*

$\alpha = 0.01$

Step -3: *Test Statistic*

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad with \quad d.f = n_1 + n_2 - 2 \text{ where } S_p = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}}$$

$$So \quad t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{(30.47 - 21.62) - (0)}{(12.89)\sqrt{\frac{1}{17} + \frac{1}{14}}} = \frac{8.85}{4.6520} = 1.90$$

Step -4: *Critical* Re *gion*

$$t > t_{\alpha(n_1+n_2-2)} \Rightarrow t > t_{(0.01)(17+14-2)} \Rightarrow t > t_{0.01(29)} \Rightarrow t > 2.462$$

Step -5: *Conclusion*

Since calculated value of t does not lie in CR so do not reject H_0.

P-VALUE APPROACH

Step -1: *Formulation of Hypotheses*

$H_0 : \mu_1 \le \mu_2$

$H_1 : \mu_1 > \mu_2$

Step -2: *Level of Significance*

$\alpha = 0.01$

Step -3: *Test Statistic*

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad with \quad d.f = n_1 + n_2 - 2 \text{ where } S_p = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}}$$

$$So \quad t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{(30.47 - 21.62) - (0)}{(12.89)\sqrt{\frac{1}{17} + \frac{1}{14}}} = \frac{8.85}{4.6520} = 1.90$$

Step -4: $P-value$

$$p = \Phi(t_0) = \Phi(1.90) = 0.034$$

Step -5: *Conclusion*

Since $p-value > \alpha$ *so do not reject* H_0.

CONFIDENCE INTERVAL APPROACH

Step -1: *Formulation of Confidence Interval*

The $(1-\alpha)100\%$ *CI for* $\mu_1 - \mu_2$ *is given by*

$$\mu_1 - \mu_2 \ge (\bar{x}_1 - \bar{x}_2) - t_{\alpha(d.f)}S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$\mu_1 - \mu_2 \ge (30.47 - 21.62) - t_{0.01(29)}(12.89)\sqrt{\frac{1}{17} + \frac{1}{14}}$$

$$\mu_1 - \mu_2 \ge (8.85) - (2.462)(12.89)(0.36)$$

$$\mu_1 - \mu_2 \ge -2.5747$$

Step -2: *Conclusion*

Since $\mu_1 - \mu_2 = 0$ *lies in CI so do not reject* H_0.

HAZBER SAMSON